Drona Raj Gyawali

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EDUCATION

Tribhuvan University Butwal, Nepal
Bachelor of Information & Communication Technology Education *Mar 2024 – Present*

Relevant Coursework: Data Structures & Algorithms, Database Management Systems, Operating Systems, ML

EXPERIENCE

Google Summer of Code Contributor with The Honeynet Project Remote
Engineering Intern *May 2026 – Sep 2026*

- Designed and implemented a **REST API for external honeypot event ingestion** in GreedyBear, enabling external sensors to submit events for IOC processing.
- Engineered an **asynchronous batch-ingestion pipeline** processing **10,000 raw honeypot entries per batch**, decoupling event validation and IOC processing from API requests through background workers.
- Implemented **sensor management APIs and data models** with source validation and sensor-level isolation to prevent asset collisions across independent ingestion sources.
- Diagnosed and resolved a **cross-container task-signing failure** between Django web and worker processes, restoring cryptographic verification for asynchronous jobs.

PROJECTS

LLM Router – AI Gateway & Semantic Router *(Python, FastAPI, NumPy, PostgreSQL, Redis)* 2026

- Engineered a schema-driven, asynchronous AI gateway to proxy multi-provider LLM requests, enabling instant provider swapping and multi-tenant budget enforcement with zero code changes.
- Reduced intent routing and safety check latency to sub-10ms by building a vector engine that computes cosine similarity via pre-normalized L_2 unit embeddings and matrix dot products, eliminating extra LLM API calls.
- Mitigated prompt injection and security risks by designing an automated dataset benchmarking pipeline to tune vector thresholds across 7 safety categories.
- Maximized uptime during upstream outages by architecting failover cascading, sub-15s resilience timeouts, and dynamic token-aware routing for long-context prompts ($>$ 8k tokens).

Annales – RAM-First Key-Value Store *(Go, MinIO, S3 API, Docker)* 2026

- Engineered a **RAM-first key-value store in Go** using immutable operation logs and an S3 CAS manifest protocol to coordinate concurrent writes and deliver **1.5ms P50 read latency**.
- Reduced log replay overhead by architecting **snapshot-based state recovery** and background log compaction, preserving crash consistency across uncommitted orphan objects.
- Benchmarked write throughput and tail latencies up to 100 concurrent clients, isolating CAS manifest contention bottlenecks to guide future multi-instance consistency designs.
- Validated system correctness by writing **13 S3-backed integration tests** covering crash recovery, snapshot replaying, concurrent commits, and uncommitted log filtering.

Quark – Multimodal RAG Engine *(Python, TypeScript, Redis, Qdrant, React.js.)* 2026

- Architected a **modular retrieval-augmented generation (RAG) pipeline** separating document ingestion, retrieval, and supporting services, with asynchronous workers for background tasks.
- Implemented **dual-stream memory** using Redis for short-term session state and Mem0 for persistent long-term context.
- Integrated **Voyage AI embeddings, Qdrant vector search, metadata filtering, and semantic reranking** to retrieve relevant context from ingested documents.

TECHNICAL SKILLS

Languages:	Python, JavaScript/TypeScript, Go, C/C++, SQL
Frameworks & Libraries:	Node.js, Django, FastAPI, React.js, Next.js, Numpy, LangChain
Tools & Platforms:	Docker, Linux, Redis, S3 API, Nginx
Engineering Concepts:	Concurrency, Asynchronous processing, Message queues, Distributed systems
Developer Tools:	Git, GitHub Actions, CMake, Pytest, Postman